Documentation

Gym Management System with Trainer, Progress Tracking, Diet & Payment Modules

Sr. No.	Details	Description
1	**Project Title**	Gym Management System
2	**Problem Statement**	• Most small and mid-sized gyms still manage memberships, attendance, fee collection and diet/workout plans manually using registers or spreadsheets. • This causes issues such as lost member records, missed payment follow-ups, no centralized way for trainers to track a member's progress, and no easy way for members to view their own diet plan, attendance or payment history. • There is a need for a centralized web-based system where the gym admin, trainers and members can interact digitally to manage memberships, plans, payments and progress efficiently.
3	**Project Objective(s)**	• To digitize and automate day-to-day gym operations (member management, attendance, fees, plans). • To allow trainers to create, update and monitor personalized diet and workout plans for each member. • To provide members a dashboard to track their fitness progress (weight, BMI, workout logs) over time. • To implement a secure online/offline payment module with receipt generation and due-date reminders. • To provide role-based access (Admin, Trainer, Member) with a clean, responsive and easy-to-use interface.
4	**Project Domain**	Web Development (Fitness / Health-Tech Management System)
5	**Project Type**	Web Application (Responsive – accessible on desktop and mobile browsers)
6	**Target Users**	• Gym Admin / Owner – manages the overall gym system. • Trainer – manages assigned members' diet, workout and progress. • Member / Customer – views own plans, progress and makes payments.
7	**User Roles**	• Admin: Add/Update/Delete members & trainers, manage plans (membership packages), view payment reports, manage announcements, full system control. • Trainer: View assigned members, create/update diet & workout plans, log member progress, mark attendance. • Member: View profile, view assigned diet/workout plan, track own progress, view payment history, make fee payment, download invoice.
8	**Major Modules**	• Authentication & Role Management Module (Admin/Trainer/Member login, registration,). • Member Management Module (Add, Update, Delete, View member profiles by Admin). • Trainer Management Module (Assign trainers to members, trainer profile management).

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		• Membership & Package Module (Monthly/Quarterly/Annual plans). • Diet & Workout Plan Module (Trainer creates/updates personalized plans). • Progress Tracking Module (Weight, BMI, body measurements, workout logs with charts). • Attendance Module (Daily check-in/check-out tracking). • Payment Module (Fee payment, due reminders, invoice/receipt generation, payment history). • Dashboard & Reports Module (Admin analytics – revenue, active members, expiring memberships).
9	**Front-End Technology**	HTML, CSS3, JavaScript, Bootstrap 5 (for responsive, modern UI), jQuery/AJAX (for dynamic without-reload updates), Chart.js (for progress graphs)
10	**Back-End Technology**	PHP , npm (for managing front-end packages such as Bootstrap, Chart.js, build tools)
11	**Database**	MySQL (managed and administered via phpMyAdmin)
12	**Tools/Software**	VS Code, XAMPP/WAMP (Apache + MySQL + PHP server stack), phpMyAdmin, Git & GitHub, Postman (API testing), Figma (UI wireframing)
13	**Hardware Requirements (if any)**	Not mandatory for the base web system. Optional (future scope): Biometric/RFID device for automated attendance, Weighing scale/BMI sensor for auto-entry of progress data.
14	**Software Requirements**	• Operating System: Windows/Linux/macOS. • Server Environment: Apache Server, PHP 8.x, MySQL 8.x. • Libraries/Frameworks: Bootstrap 5, jQuery, Chart.js, PHPMailer (for email notifications). • Browser: Any modern browser (Chrome, Edge, Firefox).
15	**Expected Features**	• Secure role-based login for Admin, Trainer and Member with session management. • Admin can Add/Update/Delete/Search members and trainers • Trainer-assigned personalized diet plan and workout schedule for every member. • Progress tracking dashboard with visual charts (weight loss/gain, BMI trend). • Attendance marking and monthly attendance report. • Online payment gateway integration (e.g., Razorpay/PayPal sandbox) plus manual/cash entry option, with auto-generated invoice/receipt (PDF). • Well-designed, mobile-responsive UI using Bootstrap with a clean dashboard for each role. • Search, sort and filter options for member and payment records. • Admin analytics dashboard – total members, active/expired memberships, monthly revenue.

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16	**References**	• Similar existing systems: Gym Master, Mindbody, Zen Planner (commercial gym management software). • Research/base references: IEEE and academic papers on "Web-based Gym/Fitness Management System". • Smart India Hackathon-style problem statements on Digital Health & Fitness Management can be reviewed on the SIH portal for additional feature ideas.